



### 3 Complex Analysis 2024 Tutorial 6

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## Consequences of uniform convergence

**Problem 24** Let  $G \subseteq \mathbb{C}$  be a region, let  $\gamma$  be a path in  $G$ , and let  $(f_n)_{n=1}^{\infty}$  be a sequence of continuous functions  $f_n : G \rightarrow \mathbb{C}$ .

- (a) Suppose that  $f_n$  converges to  $f : \gamma \rightarrow \mathbb{C}$  uniformly on  $\gamma$ . We know that  $f$  is continuous on  $\gamma$ . Prove that

$$\lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} f(z) dz.$$

- (b) Suppose that for each  $n \in \mathbb{N}$  there is  $M_n \geq 0$  such that  $|f_n(z)| \leq M_n$  for all  $z \in \gamma$ , and that the series  $\sum_{n=1}^{\infty} M_n$  converges. Prove that

$$\sum_{n=1}^{\infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} \sum_{n=1}^{\infty} f_n(z) dz.$$

Remark. This statement is used in the proof of Taylor's Theorem.

## Cauchy's integral formula

**Problem 25** Evaluate the integral

$$\frac{1}{2\pi} \int_{\gamma} \frac{e^z}{z^2 + 1} dz,$$

where

- (a)  $\gamma(t) = i + e^{it}$ ,  $t \in [0, 2\pi]$ ,  
(b)  $\gamma(t) = -i + e^{it}$ ,  $t \in [0, 2\pi]$ ,  
(c)  $\gamma(t) = 3e^{it}$ ,  $t \in [0, 2\pi]$ .

Hint: Partial fraction decomposition might help.

**Problem 26** Let  $\gamma(t) = 2e^{it}$ ,  $t \in [0, 2\pi]$ . Evaluate the following integrals.

$$(a) \int_{\gamma} \frac{z^3 + 5}{z - i} dz, \quad (b) \int_{\gamma} \frac{1}{z^2 + z + 1} dz, \quad (c) \int_{\gamma} \frac{\sin z}{z^2 - 1} dz.$$

**Problem 27** Evaluate the integral

$$\int_{\gamma} \frac{\operatorname{Re} z}{z - \frac{1}{2}} dz,$$

where  $\gamma(t) = e^{it}$ ,  $t \in [0, 2\pi]$ .

Hint: On the unit circle, the function  $\operatorname{Re} z$  coincides with a holomorphic function.